

TASK 5

Introduction

During Week 1 of the internship, I learned the fundamentals of how a software company operates. The orientation handbook provided an overview of company structure, different team types, collaboration processes, Agile methodology, Scrum framework, work breakdown techniques, sprint boards, and common software development terminology. This knowledge is essential for understanding how software products are planned, developed, tested, and delivered to customers.

Objectives

- To understand the organizational structure of a software company.
- To learn the roles and responsibilities of different teams.
- To understand how teams collaborate during software development.
- To gain knowledge of Agile methodology and Scrum practices.
- To learn how software work is planned and tracked using sprint boards.
- To become familiar with common software development terminology.

Company Structure

A software company is divided into several departments that work together to achieve business goals. Leadership is typically headed by the Chief Executive Officer (CEO) or Founder. Under the leadership team are departments such as Engineering, Product, Design, Quality Assurance (QA), and DevOps.

Key Departments

- Engineering Department – Builds and maintains software applications.
- Product Department – Decides what features should be developed.
- Design Department – Creates user-friendly and visually appealing interfaces.
- QA Department – Tests software to ensure quality and reliability.
- DevOps Department – Maintains infrastructure and deployment systems.

This structure ensures clear responsibilities while encouraging collaboration across teams.

Types of Teams

Engineering Team

The engineering team develops and maintains software products. It includes:

- Frontend Developers
- Backend Developers
- Mobile Developers
- Tech Leads

Product Team

The product team determines business requirements and feature priorities. Roles include:

- Product Manager
- Business Analyst
- Researcher

Design Team

The design team focuses on user experience and interface design. Roles include:

- UX Designer
- UI Designer
- UX Researcher

QA Team

The QA team ensures software quality through testing and validation. Roles include:

- QA Engineer
- Test Automation Engineer
- QA Lead

DevOps Team

The DevOps team manages deployment and infrastructure. Roles include:

- DevOps Engineer
- Site Reliability Engineer (SRE)

- Cloud Architect

Data Team

The data team analyzes product performance and business metrics. Roles include:

- Data Engineer
- Data Analyst
- Data Scientist

Team Collaboration

Software development involves collaboration among multiple teams. The workflow typically follows these stages:

1. Customer identifies a need or problem.
2. Product team defines requirements.
3. Design team creates wireframes and prototypes.
4. Engineering team develops the solution.
5. QA team tests the product.
6. DevOps team deploys the product to production.

This structured process ensures that customer requirements are effectively translated into working software.

Agile Methodology

Overview

Agile is a modern software development philosophy that emphasizes flexibility, continuous improvement, and customer feedback.

Waterfall vs Agile

Waterfall Approach

- Extensive planning before development.
- Limited customer feedback during development.
- Testing occurs near the end of the project.
- High risk of delays and outdated requirements.

Agile Approach

- Development occurs in short iterations called sprints.
- Frequent customer feedback is incorporated.
- Continuous testing and improvement.
- Faster delivery of working software.

Agile Values

The Agile Manifesto emphasizes:

1. Individuals and interactions over processes and tools.
2. Working software over comprehensive documentation.
3. Customer collaboration over contract negotiation.
4. Responding to change over following a fixed plan.

Scrum Framework

Scrum Roles

Product Owner (PO)

Responsible for managing the product backlog and prioritizing work according to business needs.

Scrum Master (SM)

Facilitates Scrum processes, removes obstacles, and ensures effective team collaboration.

Development Team

A cross-functional team responsible for delivering sprint goals and producing working software.

Sprint Cycle

The Scrum process follows a repeating cycle:

1. Product Backlog
2. Sprint Planning
3. Sprint Execution
4. Sprint Review

5. Sprint Retrospective

Scrum Ceremonies

Sprint Planning

Defines sprint goals and selects work items for the sprint.

Daily Standup

A short daily meeting where team members discuss:

- Work completed yesterday
- Work planned for today
- Any blockers

Sprint Review

Demonstration of completed work to stakeholders for feedback.

Retrospective

Team discussion on successes, challenges, and improvements.

Backlog Refinement

Review and preparation of future backlog items.

Work Breakdown Structure

Software projects are divided into manageable components:

Epic

A large feature or objective that spans multiple sprints.

Example:

User Authentication System

User Story

A feature described from the user's perspective.

Example:

"As a user, I want to reset my password via email so that I can regain account access."

Task

A specific implementation activity assigned to a team member.

Example:

Develop password reset API endpoint.

Bug

A defect that requires correction and prioritization.

Example:

Password reset email not delivered to Gmail users.

Story Points and Estimation

Story points estimate complexity and effort rather than time.

Common scale:

- 1 – Trivial
- 3 – Small
- 5 – Medium
- 8 – Large
- 13 – Extra Large

Velocity measures the number of story points completed during a sprint and helps predict future performance.

Sprint Board

A sprint board tracks task progress through stages:

- To Do
- In Progress
- In Review
- Done

Tools commonly used:

- Jira
- Linear
- Trello

- Asana

Sprint boards provide visibility into project progress and help teams monitor workload and deadlines.

Learning Outcomes

Through this orientation, I gained knowledge about:

- Software company organizational structure.
- Roles of different teams and departments.
- Agile development principles.
- Scrum roles, ceremonies, and sprint cycles.
- Work breakdown techniques such as epics, stories, and tasks.
- Sprint boards and project tracking methods.
- Common software engineering terminology.

Conclusion

The Week 1 orientation provided a comprehensive understanding of how software companies function and how teams collaborate to deliver products. Learning about Agile and Scrum methodologies has given me insight into modern software development practices. This knowledge will help me contribute effectively during my internship and understand the workflow of professional software development teams.